

Deep Learning for Traffic Prediction and Trend Deviation Identification: A Case Study in Hong Kong

Abstract

This paper presents a robust method for predicting traffic volume and speed on major strategy routes in Hong Kong, leveraging data from data.gov.hk and employing deep learning models. The proposed approach provides future predictions ranging from 6 minutes to one hour, considering the reliability of traffic detectors. The method involves extracting hidden deep features from the historical data of each detector to derive the detector profile. Subsequently, detectors are grouped into clusters based on the similarity of their profiles. A CNN-LSTM prediction model is provided for each cluster. The proposed approach demonstrates its resilience to detector failures. Various tests evaluating prediction performance across detector failure rates ranging from 1% to 20% validate its capability to deliver accurate predictions despite the presence of failed detectors. Without failed detectors, MAE, RMSE, and MAPE for traffic volume prediction for the next 6 minutes are 5.17 vehicles/6 minutes, 7.64 vehicles/6 minutes, and 14.07%, respectively. For traffic speed prediction are 3.70 km/h, 6.32 km/h, and 6.33%. Considering a failure rate of approximately 6% in the Hong Kong dataset, when 6% of the detectors failed, the MAE, RMSE, and MAPE for traffic volume prediction are 5.24 vehicles/6 minutes, 7.81 vehicles/6 minutes, and 14.21%, respectively. The corresponding values for traffic speed prediction are 3.87 km/h, 6.55 km/h, and 6.68%. However, the limitation of the proposed method is its potential to underperform when predicting rare or unseen scenarios, indicating the need for future research to incorporate additional data sources and methods to enhance predictive performance.

Keywords: traffic volume prediction, traffic speed prediction, deep learning, clustering method, detector profile

1 Introduction

Traffic congestion is a prominent issue in urban transportation systems, especially in densely populated cities. With its limited land and large population, Hong Kong has been dedicated to implementing various measures to optimize the transportation system and alleviate traffic congestion. One practical and cost-efficient method is intelligent traffic signal control systems, which enable dynamic adjustment of signal

phases based on real-time traffic flow, reducing congestion [1–3]. By incorporating short-term traffic variables prediction, such as traffic volume and speed, signal phases can be more accurately adjusted to adapt to near-future traffic changes. This proactive approach, such as extending the green light duration when an increase in traffic volume is predicted, effectively reduces waiting time, improves overall road capacity, and alleviates traffic pressure.

To collect data on traffic volume, speed, and occupancy, traffic detectors have been installed on all strategic routes in Hong Kong, with the installation completed by the end of 2020. Data-driven analysis and deep learning models can be used to leverage the vast traffic data collected to understand underlying traffic patterns and predict short-term traffic volume and speed [4, 5]. In recent years, deep learning (DL) methods have demonstrated significant advantages in predicting short-term traffic volume and traffic speed [6–8]. These methods excel in modeling complex nonlinear relationships. Some works focus on predicting traffic for individual stations or road sections. For instance, Xu et al. [9] proposed a hybrid model that combines autoregressive fractionally integrated moving average and nonlinear autoregressive neural network models for traffic flow prediction. Lv et al. [10] employed stacked autoencoders to predict short-term road traffic flow. Qu et al. [11] utilized a stacked recurrent neural network (RNN) to extract features from historical time series and employed an autoencoder to capture contextual features like dates and weeks to assist in traffic speed prediction. Han et al. [12] proposed DeepCluster, a method that applies k-means clustering to group all detectors and develops individual detector prediction models for each cluster. For the daily flow prediction of a single detector, Song et al. [13] divided the historical daily data into groups using density-based clustering and trained a feedforward neural network model, GMDH, for each group. Long Short-Term Memory (LSTM) models and their variants, known for effectively capturing temporal dependencies and patterns in time series data, have been widely applied in traffic prediction studies [14–16]. Abduljabbar and Dia [17], Abduljabbar et al. [18] developed LSTMs for traffic volume/speed prediction on a single highway and validated their performance on multiple highways using a public dataset. The results showcased the outstanding performance of LSTM in this task. The results also highlighted the challenges of traffic volume prediction due to its noisier nature compared to speed measurements.

However, such single-detector input-output prediction models do not consider the accessibility between adjacent road sections or stations, thus overlooking spatial dependencies. Hence, Ma et al. [19] utilized a hybrid spatial-temporal feature selection algorithm (STFSA) to identify the four monitoring points with the highest correlation to the prediction point, constructing a spatiotemporal matrix of traffic speed. This matrix is used as input features for a prediction model based on Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU) to forecast the traffic speed at a single prediction point. Numerous studies provide a prediction model outputting predictions for all monitoring points along a single road or a few connected roads. Tan et al. [20] proposed a new short-term traffic flow prediction method based on Dynamic Tensor Completion (DTC), representing traffic data as dynamic tensor patterns and estimating future traffic using weekly, daily, and close historical data under

low-rank assumptions. However, this approach is not suitable for larger-scale traffic networks due to the large size of tensors and the amount of missing data. Zhang et al. [21] decomposed traffic speed time series data into different frequency components using Ensemble Empirical Mode Decomposition (EEMD) and constructed a three-dimensional tensor $X \in R^{(n_1, n_2, n_3)}$, where n_1, n_2, n_3 represent the number of historical time slices, decomposed components, and detectors, respectively. They then employed 3D CNN for speed prediction.

Combining CNN and LSTM has emerged as a popular solution in traffic prediction [22–24]. This combination allows CNN to capture spatial dependencies and LSTM to capture temporal relationships. Cao et al. [25] proposed a CNN-LSTM model that utilized historical traffic speed from upstream and downstream and the predicted road to predict traffic speed in the Cross-Harbour Tunnel from Hong Kong Island to Kowloon. Ke et al. [26] considered the volume impact in speed prediction and used CNN with historical traffic volume and speed data to predict traffic speed. Furthermore, attention mechanisms are incorporated to capture crucial contextual information and improve predictions. Liu et al. [27] introduced attention CNN, which used a three-dimensional data matrix constructed from traffic flow, speed, and occupancy to predict traffic speed. Wu et al. [28] proposed a traffic flow prediction model that integrated an attention mechanism, CNN, and RNN. This model utilized historical traffic speed and flow data while incorporating recent traffic flow and the preceding day and week to account for the multi-periodicity of traffic flow. Zheng et al. [29] developed a Conv-LSTM module with the attention mechanism to extract spatial and short-term temporal features and a bidirectional LSTM module to capture daily and weekly cyclical features for short-term traffic flow prediction.

However, CNN is designed for fixed-size image inputs, which poses a challenge when dealing with irregular traffic data. This implies that preprocessing of traffic data is required to transform it into inputs of fixed dimensions when applying CNN for traffic prediction. Lv et al. [30] proposed a speed prediction method based on a road network embedded convolution method and RNN. They constructed a matrix that provided information about the connectivity between adjacent road sections and utilized a combination of search operations and convolution to capture spatial relationships among connected road sections. However, preprocessing becomes challenging for large-scale traffic networks and may result in information loss or introduce noise, thus impacting the model’s performance. Thus, Graph Neural Networks (GNN) have been introduced as a solution to handle irregular inputs and learn spatial relationships [31–33].

For network-level predictions, various studies construct traffic graphs based on traffic topology. For instance, Xie et al. [34] emphasized that the connectivity of road sections is the fundamental cause of traffic congestion propagation. Consequently, they represented multiple road sections as graphs based on connectivity and proposed SeqGNN to handle graph sequences for prediction. Wang et al. [35] proposed PFNet, which utilized graph embedding techniques to capture spatial relationships and employed a Deep Multi-View Sequence Encoder (DMVSE) to capture time correlation from monthly periodicity, weekly periodicity, and closeness. The spatial correlations among different traffic network nodes are local and non-local. To address

this, Fang et al. [36] proposed the Global Spatial-Temporal Network (GSTNet), which considers both local spatial correlations among adjacent stations and global spatial correlations. Lv et al. [37] considered road connectivity and incorporated historical traffic pattern correlations and local area functionality similarity to construct traffic graphs. They proposed a traffic flow prediction model based on multi-graph convolution and GRU. Moreover, the interaction between road sections is influenced by factors such as the natural environment and socio-economic factors. Xia et al. [38] extracted static features such as spatial distance and dynamic features from the most recent traffic data using a multi-head attention network and dynamic node embedding to capture these dynamics. They integrated dynamic graph convolution modules and LSTM to achieve accurate traffic flow prediction. Chen and Chen [39] introduced a graph learning model based on reinforcement learning. They initially constructed a traffic graph based on the accessibility of road sections. Then, they utilized reinforcement learning to extract nonlinear spatiotemporal dependencies between stations, adaptively generating the graph adjacency matrix. However, GNN-based approaches need to address challenges such as handling large-scale graph structures, high-dimensional features, and the issue of gradient vanishing during information propagation. The computational cost of processing large-scale graph structures is high and may not satisfy real-time requirements.

To predict the traffic volume and speed for detectors on major strategy routes in Hong Kong, it is essential to consider the detectors' reliability since the average failure rate in Hong Kong is 6%. Handling the resulting missing data is crucial. This paper proposes an approach that utilizes data from data.gov.hk to develop a robust traffic volume/speed prediction model that is resilient to detector failures. The hidden deep features within each detector's historical daily traffic volume are extracted to obtain the profile for the corresponding detector. The detectors are divided into multiple clusters based on the similarity of detectors' profiles. For each cluster, a prediction model based on CNN-LSTM is developed to forecast the traffic volume/speed for all detectors within the cluster. Furthermore, the extracted profiles can be used to identify abnormal daily traffic patterns for each detector. This information can be utilized to investigate potential equipment issues or possibly unrecorded traffic incidents.

The remaining sections of this paper are organized as follows. Section 2 presents the proposed method and DL model architectures. Section 3 discusses the experimental results. Finally, Section 4 provides a summary of this paper.

2 Data

2.1 Description of the Data Source

The traffic data used in this study is sourced from data.gov.hk and extracted from fixed-location traffic detectors. The dataset includes timestamps, detector IDs, traffic volume, speed, and occupancy. These detectors are strategically placed on major roads and routes in Hong Kong. A total of 1,912 lane detectors corresponding to 714 sites are surveyed. The locations of these sites are depicted in Fig. 1. By examining the traffic flow-speed graph of the dataset, noteworthy changes in the relationships of

most detectors are observed in May 2022, which are attributed to detector recalibration. Consequently, data from May 2022 to November 2022 are selected for training, validation, and testing. Each time slice in this paper represents a 6-minute time interval.

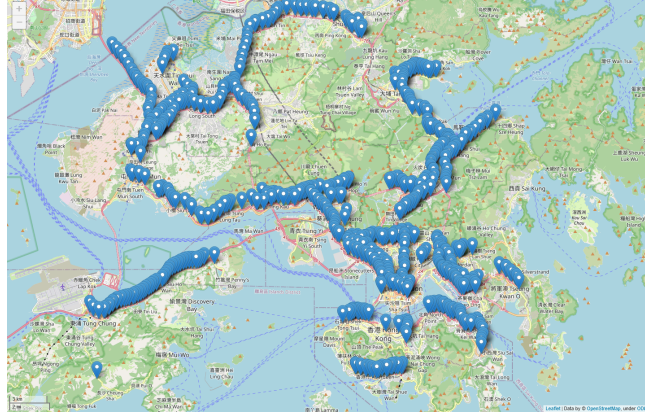


Fig. 1 Location of investigated traffic detectors in HK

Due to communication interruptions or detector failures, the collected raw traffic flow data unavoidably contains missing values. Fig. 2 displays the working status of the detectors, where blank points indicate periods when the detector was not working. The overall average failure rate of the dataset is 5.94%. Data imputation is necessary for this dataset to facilitate the training of prediction models.

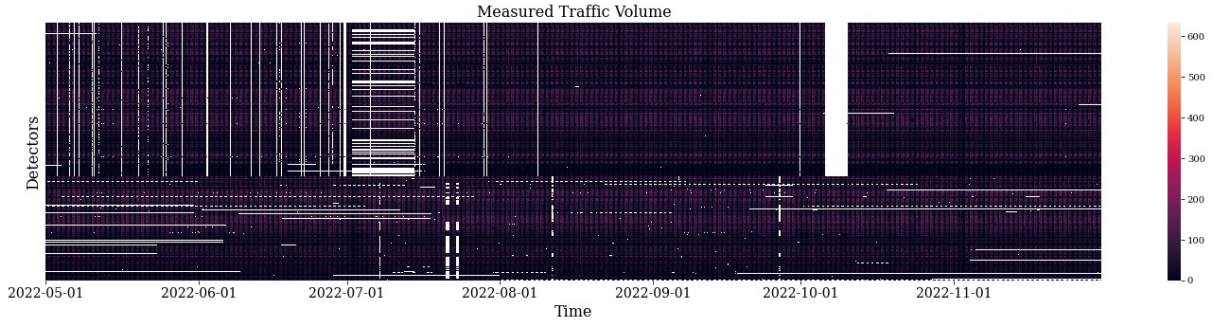


Fig. 2 Working status of traffic detectors

2.2 Data Imputation

Fig. 2 provides insights into the types of missing data, which can be categorized into three groups based on the duration of continuous detector failures. Each category requires a different data imputation method:

(1) For cases where a detector’s continuous failure duration is no more than 6 minutes, the most recent observed traffic value can be used for imputation.

(2) For cases where a detector’s continuous failure duration is more than 6 minutes but less than 6 hours, the K-Nearest Neighbors (KNN) method can be employed to find the most similar detectors on the same day. The missing traffic variables for the failed detector can be filled with the average values from similar working detectors in the corresponding time slice.

(3) For cases where a detector’s continuous failure duration exceeds 6 hours, using KNN to find similar detectors may not be accurate. In such situations, the approach involves identifying the K most similar dates for data imputation. The distance between two dates is determined by the average Euclidean distance of observed values provided by working detectors within their corresponding time slices. If the number of points with values between two dates is less than 60% of the total number of points in a day, the distance between these two dates is set to infinity. The missing values can be filled with the average values from the K nearest dates whose distances are smaller than a threshold value.

2.3 Data Normalization

To expedite the search for the optimal solution for the DL models, perform min-max normalization to scale the data to the range of $[0, 1]$:

$$X^* = \frac{X - x_{min}}{x_{max} - x_{min}}, \quad (1)$$

where X represents the input traffic variable, and x_{min} and x_{max} represent the minimum and maximum values of that traffic variable in the training dataset.

For the model’s output, perform the following operation to restore it to the original scale:

$$Y' = Y^* \times (x_{max} - x_{min}) + x_{min}, \quad (2)$$

where Y^* is the output of the model, and Y' is the traffic variable prediction after being restored to the original scale.

3 Methodology

3.1 Overview of the Proposed Method

Taking traffic volume prediction as an example, the proposed method consists of the following steps:

(1) Extracting deep features with detector signatures from each daily traffic volume sequence.

(2) Generating representative profiles for each detector based on their daily features and grouping the detectors based on profile similarity.

(3) Developing a traffic volume prediction model for each group.

By following these steps, the detectors across the Intelligent Road Network (IRN) are divided into multiple groups, each corresponding to a prediction model that operates independently of the other groups. This division allows each prediction model to handle inputs from only a portion of the entire network, reducing input dimensionality and the required network complexity, leading to faster prediction speeds. Additionally, the detectors within each group can mutually benefit from one another. When a detector fails, real-time imputation and effective prediction can be accomplished by utilizing the real-time information from other detectors within the same group. Only a few prediction models are needed for the entire traffic network, reducing the development and maintenance costs accordingly.

In the following sections, we will comprehensively explain the three steps.

3.2 Daily Feature Extraction Model

Traffic volume measured by each detector exhibits periodic changes, such as similar peak and off-peak periods on working days. However, the similarity among different locations extends beyond factors such as points of interest (POI), geographical location, and functionality. To capture these similarities, we analyze each detector's daily trends in historical traffic volume. DL models possess strong pattern analysis capabilities and are utilized to extract features from the daily traffic volumes of each detector.

The proposed feature extraction model, as illustrated in Fig. 3, consists of several CNN layers and an FC layer. It takes a daily traffic volume data of a detector $v_{x_i}^{d_j}$ as input and outputs a distribution $\hat{L}_{x_i}^{d_j}$ representing the ID of the detector corresponding to the input daily data. CNN applies convolutional operations with kernels of a certain size to local regions of the input data, computing the inner product between different data windows and the convolutional kernel to capture local relationships. Hence, CNNs are utilized to extract features from time series and reduce feature dimensionality. Initially, we perform convolution operations on the input using kernels of different sizes while maintaining the output feature map size through padding. In Fig. 3, a, b, and c represent kernel sizes of 3, 5, and 7, respectively. The convolution results are then summed together to achieve multi-scale feature extraction and fusion. Subsequently, several stacked convolutional layers are used, with the sliding stride of the convolutional kernels set to $(a - 2)$. By sliding at different positions, the convolutional kernels can traverse the entire input. The stride setting reduces the output size of the convolutional layer by half, decreasing the feature map size in each convolutional layer. After passing through multiple CNN layers and flattening, the features $y_{x_i}^{d_j}$ are obtained, representing the features of detector x_i throughout day d_j . Next, through a fully-connected layer, $y_{x_i}^{d_j}$ is mapped to an output vector $\hat{L}_{x_i}^{d_j}$, with dimensions equal to the number of detectors. Each dimension of $\hat{L}_{x_i}^{d_j}$ represents the relevance to the corresponding detector ID.

$$y_{x_i}^{d_j}, \hat{L}_{x_i}^{d_j} = f_{CLU}(v_{x_i}^{d_j}), \quad (3)$$

where f_{CLU} indicate the proposed daily feature extraction model. $v_{x_i}^{d_j}$ is the traffic volume sequence of detector x_i on day d_j , encompassing the entire day.

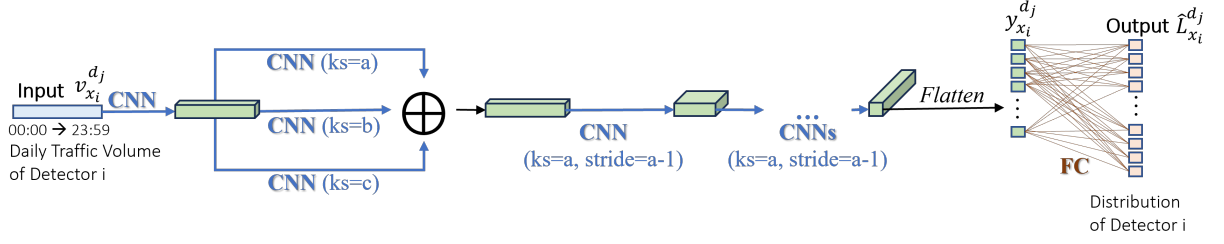


Fig. 3 Proposed daily feature extraction model

3.3 Profile Obtain

Deep features associated with each detector are collected. However, certain days may be associated with atypical traffic conditions, resulting in the deep features from these specific days potentially failing to identify the corresponding detector ID's distribution accurately. The profile is computed as the average value of the deep features that correctly identify the detector ID to make the profile a more precise representation of the detector's general characteristics.

For detector x_i on the j^{th} day, the predicted distribution $\hat{L}_{x_i}^{d_j}$ is as follows:

$$\hat{L}_{x_i}^{d_j} = \{p_{x_1}^{d_j}, p_{x_2}^{d_j}, \dots, p_{x_i}^{d_j}, \dots, p_{x_N}^{d_j}\} \quad (4)$$

where $p_{x_i}^{d_j}$ represents the model's estimation of the probability that the input daily data belongs to the i^{th} detector, and N is the total number of detector IDs.

Sort $\hat{L}_{x_i}^{d_j}$ in descending order based on probabilities, resulting in:

$$\hat{L}'_{x_i}^{d_j} = \{p_{x_{i_1}}^{d_j}, p_{x_{i_2}}^{d_j}, \dots, p_{x_{i_N}}^{d_j}\} (p_{x_{i_1}}^{d_j} \geq p_{x_{i_2}}^{d_j} \geq \dots \geq p_{x_{i_N}}^{d_j}) \quad (5)$$

Denote the top 1% of the sorted probabilities in $\hat{L}'_{x_i}^{d_j}$ as $P'^{d_j}_{x_i} = \{p_{x_{i_1}}^{d_j}, p_{x_{i_2}}^{d_j}, \dots, p_{x_{i_{N/100}}}^{d_j}\}$

$$\bar{y}_{x_i} = \frac{1}{k} \sum_{j=1}^k y_{x_i}^{d_j}, \text{ if } p_{x_i}^{d_j} \in P'^{d_j}_{x_i}, \quad (6)$$

where $\bar{y}_{x_i}$ is the obtained profile of detector x_i .

The Euclidean distance between each pair of detector profiles is computed, and the hierarchical clustering algorithm f_{HA} is used to divide the detectors into multiple clusters ($C_1, C_2, \dots, C_M$):

$$\begin{aligned} C_1, C_2, \dots, C_M &= f_{HA}(\bar{y}_{x_1}, \bar{y}_{x_2}, \dots, \bar{y}_{x_N}) \\ C_1 \cup C_2 \cup \dots \cup C_M &= \{x_1, x_2, \dots, x_N\} \end{aligned} \quad (7)$$

3.4 Proposed Traffic Volume Prediction Model Architecture

After clustering, each cluster is composed of detectors with similar characteristics. A separate traffic volume prediction model is constructed for each cluster, as illustrated

in Fig. 4. For a given Cluster $C_J (J = 1, 2, \dots, M)$, the LSTM and CNN are employed to extract spatial-temporal features. The LSTM captures global information for the cluster, while the CNN utilizes the individual detector's time series to obtain each detector's deep features separately. The obtained deep features are then fused to generate traffic volume predictions for the cluster for the next hour. For example, for Cluster C_J composed of n detectors $(x_{J_1}, x_{J_2}, \dots, x_{J_n})$:

$$C_J = \{x_{J_1}, x_{J_2}, \dots, x_{J_n}\}, v_{C_J}^{\tau-\Delta} = \{v_{x_{J_1}}^{\tau-\Delta}, v_{x_{J_2}}^{\tau-\Delta}, \dots, v_{x_{J_n}}^{\tau-\Delta}\} \quad (8)$$

$$y_{C_J}^\tau = f_{FC} f_{LSTM} f_{FC}(v_{C_J}^{\tau-10\Delta}, v_{C_J}^{\tau-9\Delta}, \dots, v_{C_J}^{\tau-\Delta}) \oplus f_{CNNs}(v_{C_J}^{\tau-10\Delta}, v_{C_J}^{\tau-9\Delta}, \dots, v_{C_J}^{\tau-\Delta}) \quad (9)$$

$$(\hat{v}_{C_J}^\tau, \hat{v}_{C_J}^{\tau+\Delta}, \dots, \hat{v}_{C_J}^{\tau+9\Delta}) = f_{CNNs}(y_{C_J}^\tau) \quad (10)$$

where $y_{C_J}^\tau$ represents the concatenated deep features extracted from the cluster using the two neural networks. $\hat{v}_{C_J}^\tau$ is an n -dimensional vector representing the predicted traffic volume at time τ for each detector in the cluster.

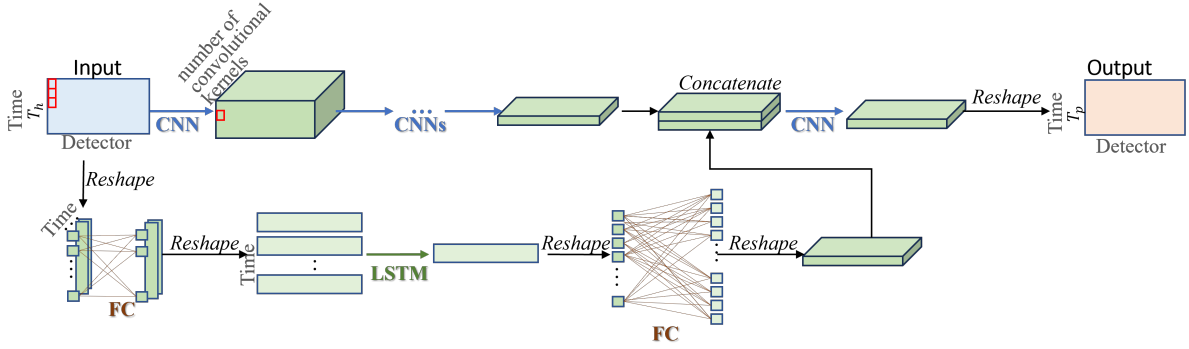


Fig. 4 Proposed DL traffic volume prediction model

In Eq. (10), f_{CNNs} utilizes multiple convolutional layers to extract temporal features from the input data. At each layer, neighbouring k_t^l features are combined along the temporal dimension. The convolutional kernels at each layer have a size of $(k_t^l, 1)$, where 1 indicates that no feature fusion is performed in the spatial (detector) dimension. By passing through multiple convolutional layers, the temporal dimension is compressed from the input T_h to 1. Since the desired output is the prediction for the next T_p time steps, the number of convolutional kernels in the final layer is set to T_p . The resulting output from the right-hand side of the equation is a tensor of size $(T_p, 1, N)$.

On the left-hand side of Eq. (10), cluster information is extracted. Firstly, for each time step in the data, spatial features are extracted using a fully connected layer

(f_{FC}). The input information from N detectors at each time step is connected to a new set of N units. The information from different time steps is independent of each other. Subsequently, the new N -dimensional features from each time step are fed into an LSTM (f_{LSTM}) to extract temporal features and obtain the feature prediction for the next time step, resulting in a tensor with the size of $(1, N)$. Afterward, a fully connected layer is used to obtain a $(T_P \times N)$ -dimensional vector, which is then transformed into a tensor of size $(T_P, 1, N)$.

Finally, feature concatenation is performed. Each detector at each time step has a 2-dimensional feature. The final prediction is obtained using a convolutional layer with a kernel size of $(2, 1)$. The proposed prediction model not only considers the temporal features of each detector but also incorporates cluster information to assist in the prediction process.

3.5 Training of DL Models

The daily feature extraction model, which functions as a multi-classification neural network, utilizes the cross-entropy loss function to minimize the disparity between the predicted probability distribution and the one-hot encoded representation of the actual detector ID. The input daily traffic volume is standardized to ensure uniformity. Model parameters are updated through backpropagation and gradient descent optimization. The maximum number of training iterations for the model is capped at 300.

Regarding the Traffic Volume Prediction Model, the mean squared error (MSE) loss function measures the discrepancy between the predicted traffic volume and the actual values. During training, the standardized historical 1-hour traffic volume data for each detector within a cluster is used as input, while the corresponding standardized ground truth values (i.e., the future 1-hour traffic volume) serve as the target for prediction.

All DL models are trained iteratively on a training dataset, and the performance is evaluated on a validation dataset. Hyperparameters, including learning rate, batch size, and network architecture, are fine-tuned for optimal performance. Adam optimization is employed as the optimizer for all deep models. The learning rate, a critical parameter affecting convergence, is selected through multiple experiments within the range of $[10^{-3}, 10^{-4}, 10^{-5}]$. A dynamic learning rate scheme is implemented, reducing the learning rate by a factor of 0.1 if the model's evaluation metric fails to improve after 30 training iterations until it reaches the minimum value of 10^{-6} . Additionally, an early stopping strategy is utilized to prevent overfitting.

4 Results

The proposed method involved training and validating traffic volume and speed prediction models using datasets collected from major roads in Hong Kong between May 2022 and October 2022. Subsequently, we evaluate the proposed method using the corresponding traffic volume and speed datasets from November 2022.

The testing included the following aspects:

- (1) Evaluating the predictive performance of the prediction models when all detectors are working.

(2) Simulating 1%-20% detector failures to assess the resilience of the prediction models to detector failures.

(3) Assessment of the proposed method's performance in offline data imputation for addressing missing data in the dataset.

(4) Additionally, we evaluated the accuracy of the proposed daily feature extraction model in identifying detector IDs.

These tests enabled a comprehensive understanding of the proposed method's performance in various scenarios, including its predictive capabilities under both normal and detector failure conditions, its performance in data imputation, and the accuracy of the daily feature extraction model in identifying detector IDs.

4.1 Evaluation Metrics Used for Testing

The predictive model is evaluated using the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

$$\text{MAE}(Y, \hat{Y}) = \frac{1}{N} \sum_{i=1}^N |Y_i - \hat{Y}_i|, \quad (11)$$

$$\text{RMSE}(Y, \hat{Y}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2}, \quad (12)$$

$$\text{MAPE}(Y, \hat{Y}) = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{Y}_i - Y_i|}{Y_i}, \quad (13)$$

where $\hat{Y}$ and Y represent the prediction and the observation, respectively. The predictive performance improves as MAE, RMSE, and MAPE decrease. All MAPE calculations for traffic volume/speed prediction only consider observations with a traffic volume greater than 10 vehicles per 6 minutes or 1 km per hour to avoid division by zero and biases in error for low observations. For example, if the observed traffic volume is four vehicles and the predicted traffic volume is two, the MAPE is 50%, although the error is only two vehicles.

4.2 Predictive Performance of the Prediction Model without Failed Detectors

4.2.1 Traffic volume prediction without failures

The proposed method enables a one-step traffic volume prediction for the next 6 minutes to one hour. Fig. 5 presents the MAE, RMSE, and MAPE corresponding to each time interval. It also includes the MAE, RMSE, and MAPE obtained using the most recent observed traffic volume as a prediction ("Naïve"). For the 6-minute

ahead prediction, the proposed method achieves MAE, RMSE, and MAPE values of 5.17 vehicles/6 minutes, 7.64 vehicles/6 minutes, and 14.07%, respectively. Regarding the traffic volume prediction for the next hour, the obtained MAE, RMSE, and MAPE are 5.69 veh/6min, 8.96 veh/6min, and 15.26%. It can be observed that the proposed method outperforms the Naïve Method for predictions ranging from the next 6 minutes to 1 hour, with its advantage becoming more pronounced over time. Although the predictions for more extended time horizons are less accurate than the near-term predictions, the results obtained one hour ahead are still superior to using the observed traffic volume from the previous 6 minutes as a direct prediction. Fig. 6 demonstrates that the proposed method captures the trend of traffic volume changes but may underestimate traffic volume in congested conditions.

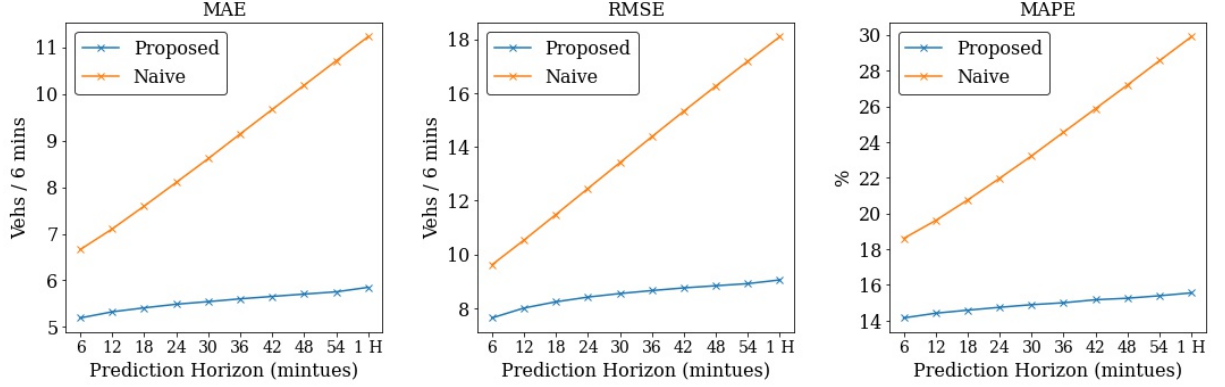


Fig. 5 Traffic volume predictive performance without detector failures

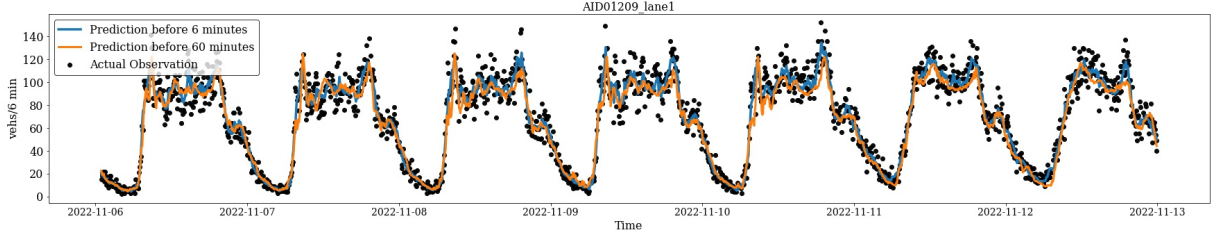


Fig. 6 Visualization of predicted traffic volume and actual observation

4.2.2 Traffic speed prediction without failures

By changing the input of the prediction model from traffic volume data to traffic speed data and retraining the prediction models, traffic speed prediction models are obtained. The evaluation metrics, including MAE, RMSE, and MAPE, are depicted in Fig. 7, showcasing the speed predictions generated by the proposed approach for the

next 6 minutes to 1 hour. The 6-minute predictions resulted in MAE/RMSE/MAPE values of 3.70 km/h, 6.32 km/h, and 6.33%, respectively. Similarly, for the subsequent 1 hour, the corresponding values are 4.24 km/h, 7.37 km/h, and 8.44% for MAE/RMSE/MAPE.

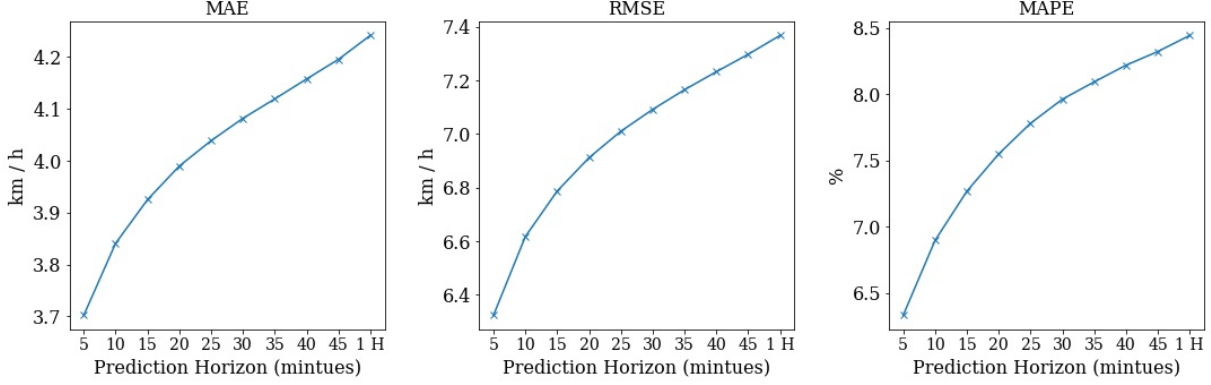


Fig. 7 Traffic speed predictive performance without detector failures

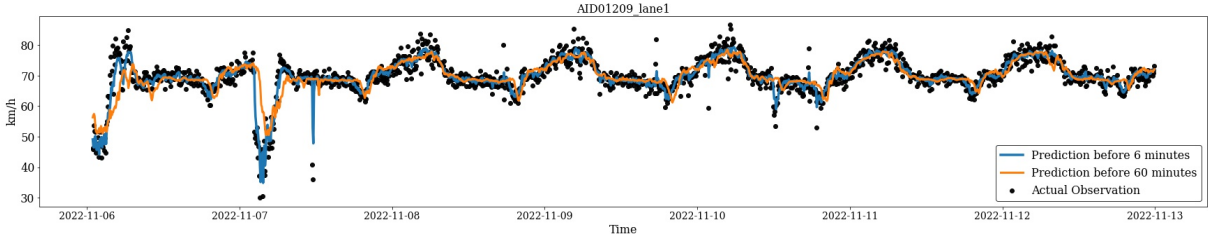


Fig. 8 Visualization of predicted traffic speed and actual observation

Table 1 Traffic Speed Prediction performance of different speed ranges

Speed Range	Proportion	MAE	RMSE	MAPE
$0 \leq x < 15$ km/h	0.58%	9.9 km/h	15.6 km/h	149%
$15 \leq x < 30$ km/h	1.42%	7.3 km/h	11.8 km/h	33%
$x \geq 30$ km/h	98%	3.7 km/h	6.1 km/h	5%

Fig. 8 provides a clear visual comparison between the predicted speeds for the next 6 minutes and 1 hour obtained using the proposed method and the actual measurements. It can be observed that the proposed method performs well in predicting traffic speeds under normal traffic conditions. It is also capable of capturing trends of sudden drops in traffic speed.

However, this method is unsuitable for extreme or rare low-speed traffic scenarios. Table 1 presents the sample proportions within different speed ranges and their respective MAE, RMSE, and MAPE values. Due to the scarcity of available samples for learning, the proposed model is unsuitable for non-routine congestion scenarios with low traffic speeds. New methods for predicting non-routine congestion speeds are needed.

Table 1 shows that the proposed method yields relatively large values for all the metrics when the speed is less than 15 km/h, indicating a significant distance between the predictions and the actual observed values. Hence, we visualized such data, as shown in Fig. 9. The ‘Median’ in the figure represents the median values of the training dataset for each corresponding time slot from Monday to Sunday. ‘Mean’ indicates the mean values. The blue line represents the visualization of the actual observed values for one week from the test set of this detector, while the orange line represents the predictions obtained from the proposed method. It can be observed that the actual observed values exhibit significant fluctuations, while the predictions are only able to fit a general trend. However, such substantial fluctuations in speed observations throughout the week are highly abnormal and likely caused by erroneous reports due to detector malfunctions. Hence, we removed these exceptional values where the difference between the current speed and the speed measured in the previous time slot exceeds 30 km/h, with a frequency greater than 50% within a day.

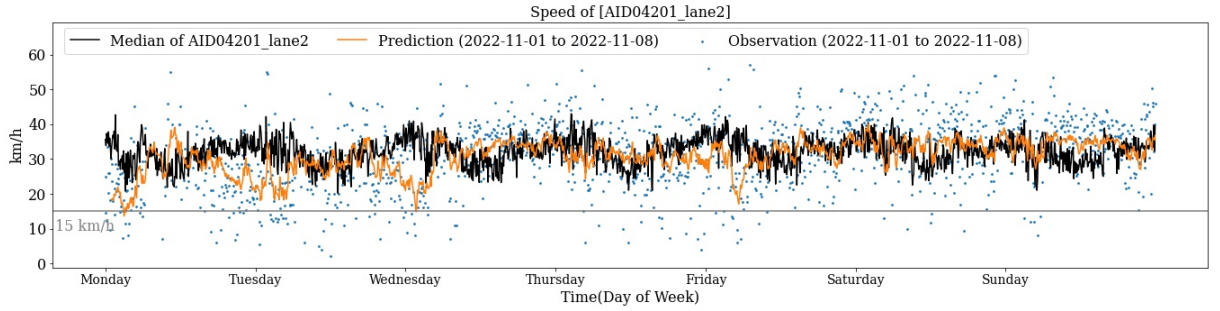


Fig. 9 Visualization of predicted traffic speed and actual observation (speed < 15 km/h) (Abnormal Speed)

Table 2 Traffic Speed Prediction performance of different speed ranges after removing abnormal speed

Speed Range	MAE	RMSE	MAPE
$0 \leq x < 15$ km/h	6.8 km/h	10.5 km/h	86%
$15 \leq x < 30$ km/h	6.4 km/h	9.7 km/h	29%
$x \geq 30$ km/h	3.3 km/h	5.0 km/h	5%

The predictive performance after removing these exceptional values is shown in Table 2. It can be observed that the predictive performance for low-speed traffic

scenarios improves after the removal. However, it is still not particularly satisfactory. We conducted visualizations to investigate possible reasons, as shown in Fig. 10. It can be observed that the variation trend of traffic speed for this detector has changed, differing from the training data used to train the prediction model. The results also highlight that DL models rely on the data used for learning. If traffic trends change, it is necessary to train the prediction model using new data to adapt to the new circumstances.

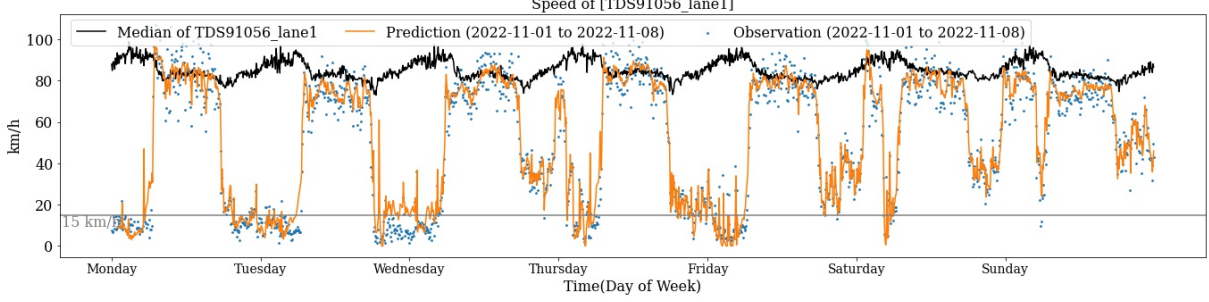


Fig. 10 Visualization of predicted traffic speed and actual observation (speed < 15 km/h) (Pattern Changed)

4.3 Predictive Performance of the Prediction Model with Detector Failures

4.3.1 Traffic volume prediction with failures

The proposed method can generate accurate traffic volume predictions even with detector failures. We conducted simulations with 1% to 20% of detectors experiencing failures to assess the method's failure resilience.

When a detector fails, the model's input contains missing values corresponding to the failed detectors. We filled in the missing inputs to make predictions based on the current available observations of other working detectors within the same cluster as the failed detector and their historical average weekly traffic volume (AWT) derived from the training dataset.

For a detector x_j in cluster C_J , if it fails at time τ (corresponding to day of the week D_a and time of the day T_b), the imputed value $\tilde{v}_{x_j}^\tau$ is obtained as follows:

$$\tilde{v}_{x_j}^\tau = AWT_{x_j}^{D_a, T_b} \times Avg_{x_l \in C_J} \frac{v_{x_l}^\tau}{AWT_{x_l}^{D_a, T_b}}, \quad (14)$$

where $\tilde{v}_{x_j}^\tau$ represents the real-time measured traffic volume of the working detector x_l in the cluster. Avg refers to the average function.

The average values of MAE/RMSE/MAPE obtained are shown in Fig. 11. Each data point in the figure represents the average of 30 experiments, where different detectors are randomly selected as failures. The results in Fig. 11 indicate a gradual decline in the model's performance as the number of failed detectors increases, which is

expected. The results show that the proposed method outperforms the Naive method in scenarios both with and without failed detectors. The degradation in performance of the proposed method is minimal. For the next 6-minute prediction, when the failure rate of detectors reaches 20%, the average degradation in performance results in an increase of 0.29 vehs/6min in MAE, 0.72 vehs/6min in RMSE, and 0.56% in MAPE (as shown in the bottom line of Fig. 11). For the prediction of the next 1 hour (as shown in the top line of Fig. 11), the degradation leads to an average increase of 0.11 vehs/6min in MAE, 0.27 vehs/6min in RMSE, and 0.21% in MAPE.

As a reference, the average failure rate in the dataset is 5.94%. With 6% failed detectors, the average MAE, RMSE, and MAPE obtained for the 6-minute prediction are 5.24 vehs/6min, 7.81 vehs/6min, and 14.21%, respectively. Compared to the scenario without detector failures, this represents increases of 0.07 vehs/6min, 0.17 vehs/6min, and 0.14%, respectively. For the 1-hour prediction with 6% failed detectors, the average MAE, RMSE, and MAPE are 5.70 vehs/6min, 8.99 vehs/6min, and 15.29%, respectively. Compared to the scenario without detector failures, this represents increases of 0.02 vehs/6min, 0.03 vehs/6min, and 0.03%, respectively.

Additionally, Fig. 12 presents a box plot illustrating the results obtained for all test cases, providing a comprehensive visualization of the proposed method's predictive performance involving different failed detectors. Even with the same failure rate, different failed detectors correspond to varying predictive performance. Nonetheless, the degradation in prediction accuracy remains within an acceptable range. Overall, the proposed method demonstrates resilience to detector failures.

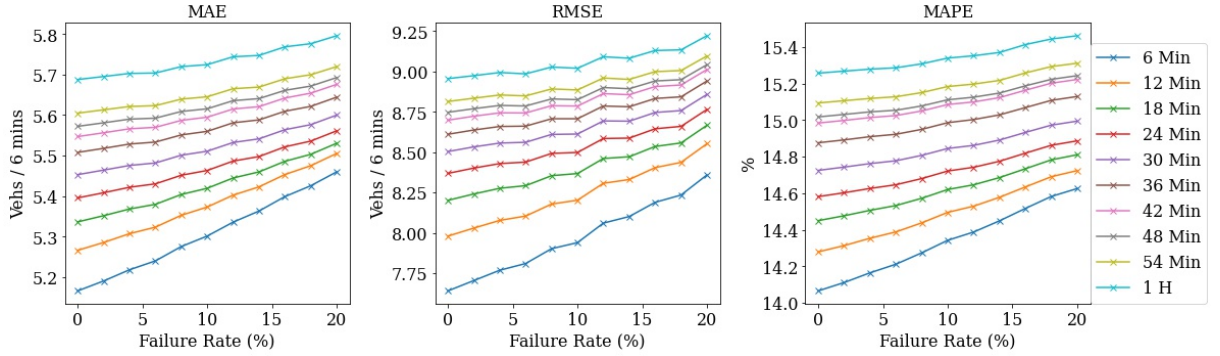


Fig. 11 Traffic volume predictive performance with detector failures

4.3.2 Traffic speed prediction with failures

The performance of traffic speed prediction with failed detectors is shown in Fig. 13. For the 6-minute predictions, when there is a 6% failure rate in the detectors, the MAE, RMSE, and MAPE values are 3.87 km/h, 6.55 km/h, and 6.68%, respectively. Compared to the predictions without failures, the average performance decreases by 0.17 km/h, 0.23 km/h, and 0.34% in MAE, RMSE, and MAPE, respectively. Fig. 14 shows box plots illustrating the MAE, RMSE, and MAPE values for each failure case

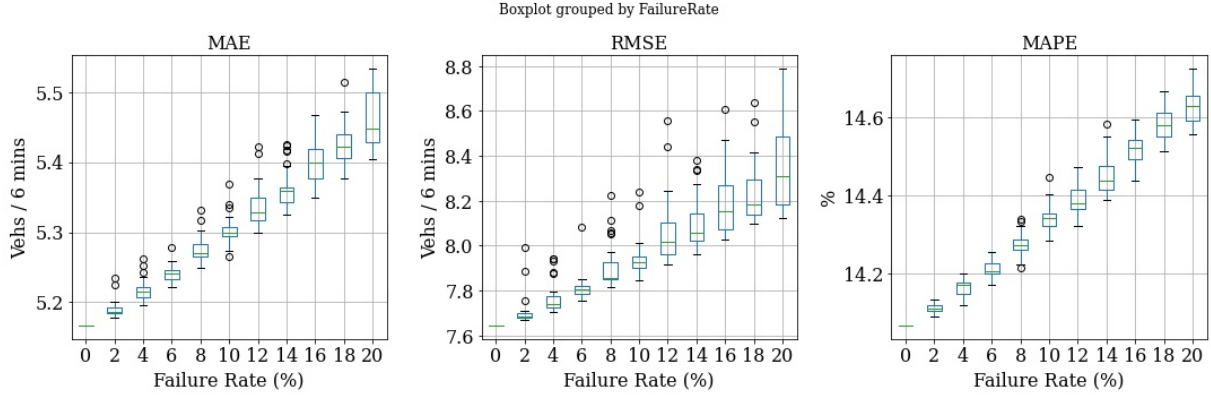


Fig. 12 Traffic volume predictive performance of each case for next 6 minutes with detector failures

in the next 6 minutes. For the predictions of the next 1 hour, with a 6% failure rate in the detectors, the MAE, RMSE, and MAPE values are 4.30 km/h, 7.43 km/h, and 8.55%, respectively. On average, there is a decrease of 0.06 km/h, 0.06 km/h, and 0.10% in MAE, RMSE, and MAPE, respectively, compared to the predictions without failures.

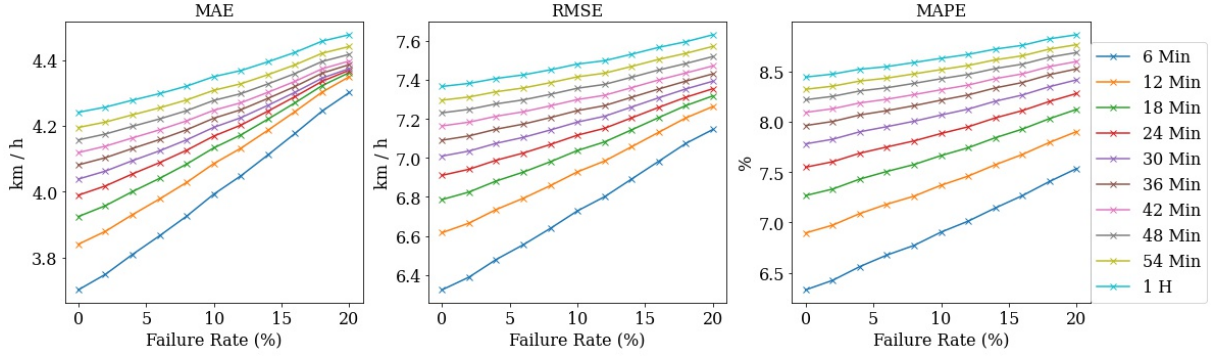


Fig. 13 Traffic speed predictive performance with detector failures

4.4 Imputation Performance of the Prediction Model for Offline Dataset

4.4.1 Traffic volume imputation

Furthermore, the proposed prediction model can be utilized for offline missing data imputation. The results of utilizing the proposed method for data imputation are presented in Fig. 15. Each data point in the figure represents the average of 30 experiments, with different detectors randomly selected as targets for imputation in each experiment. The results demonstrate that the proposed method's imputation yields

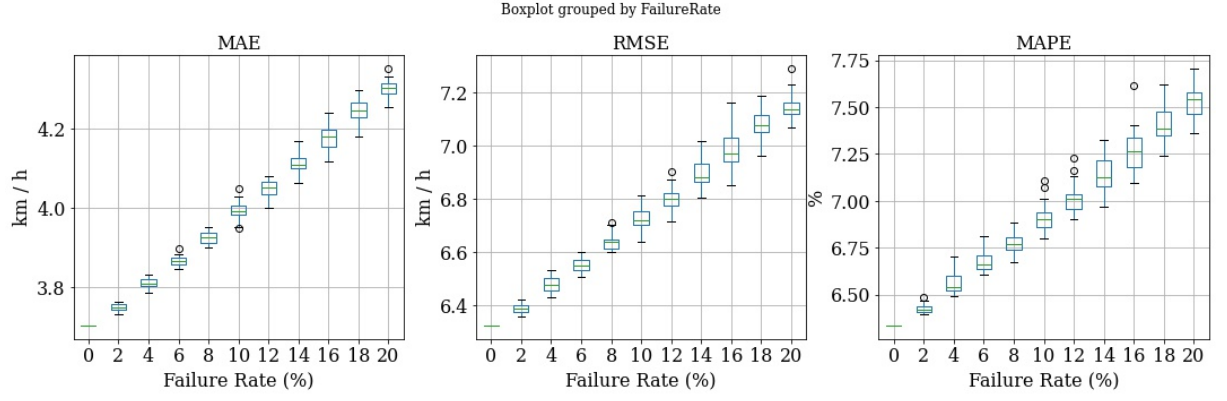


Fig. 14 Traffic speed predictive performance of each case for next 6 minutes with detector failures

MAE, RMSE, and MAPE values ranging from approximately 5.1 to 5.3 vehs/6min, 7.6 to 7.8 vehs/6min, and 14% to 14.2%, respectively. These findings underscore the effectiveness of the proposed method in accurately filling in missing data.

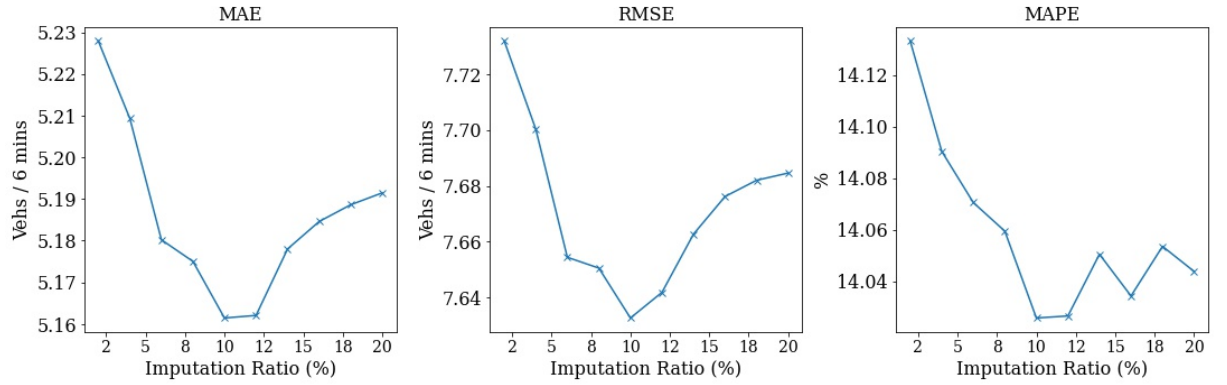


Fig. 15 Performance of offline traffic volume data imputation

4.4.2 Traffic speed imputation

The offline imputation performance of traffic speed data is depicted in Fig. 16. The results demonstrate that the imputation method yields a range of MAE, RMSE, and MAPE values ranging from 3.7 to 3.8 km/h, 6.2 to 6.4 km/h, and 6.5% to 6.9%, respectively. These findings emphasize the capability of the proposed method in effectively filling missing traffic speed data.

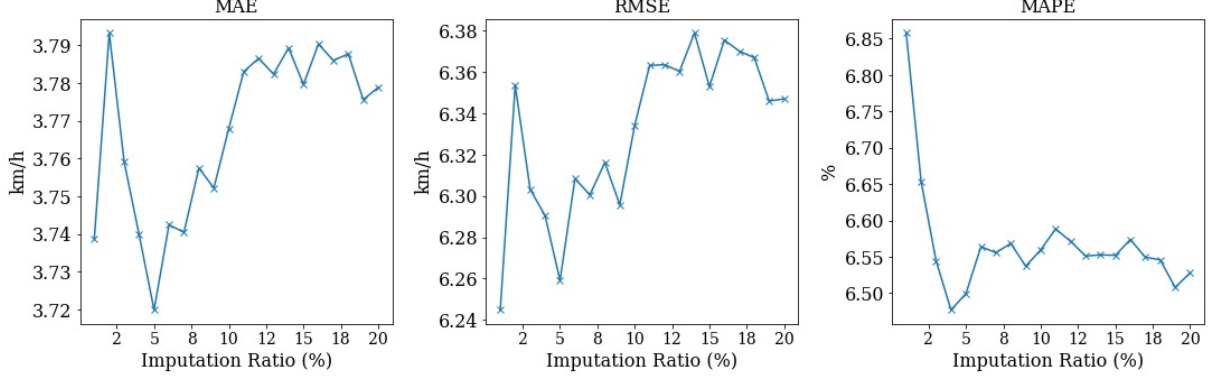


Fig. 16 Performance of offline traffic speed data imputation

4.5 Recognition Performance of the Daily Feature Extraction Model for Detector IDs

The proposed daily feature extraction model can identify the detector ID corresponding to each daily traffic volume in the dataset. The proposed daily feature extraction model produces a probability distribution, with each element indicating the likelihood of the input daily volume originating from a specific detector. As a result, it enables the calculation of the probability of each detector being correctly identified daily, as well as the percentage of all detectors that can be accurately recognized each day.

We can rank the detectors based on the probabilities provided by the model's output. In Figs. 17 and 18, the label 'Top 1' indicates that the detector ID with the highest probability predicted by the model is the correct ID. Similarly, 'Top 1%' signifies that the correct ID corresponding to the input's daily volume is among the top 1% of the ranked detectors. It should be noted that different detectors may exhibit similar profiles, thereby providing a basis for clustering them based on their similarities. Consequently, our focus lies on the top 1% ranking results.

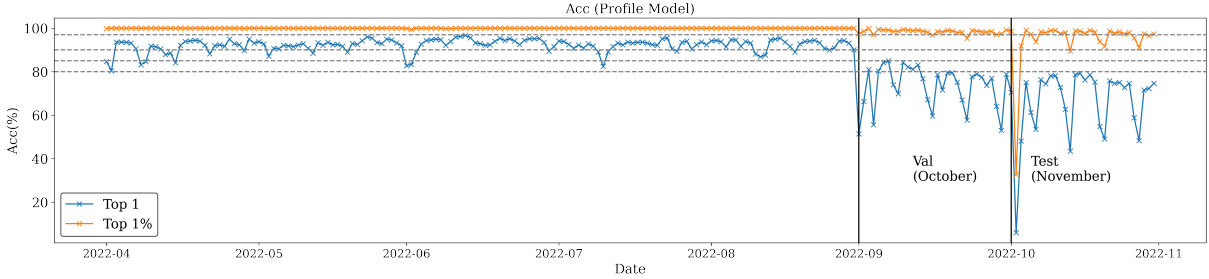


Fig. 17 Identification accuracy of each day of proposed DL daily feature extraction model

Fig. 17 illustrates that the daily feature extraction model achieves a correct identification rate of over 95% for most days. Moreover, Fig. 18 reveals that only a small

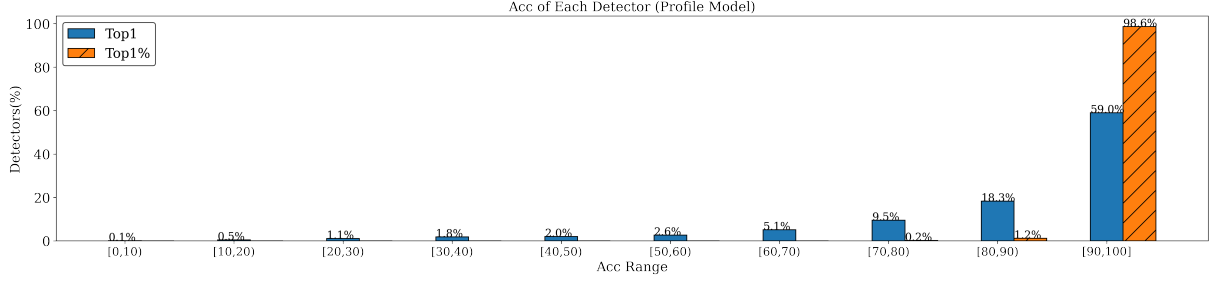


Fig. 18 Identification accuracy of each detector of proposed DL daily feature extraction model

percentage, 1.4%, of detectors have an identification accuracy below 90% but still above 70%.

The daily feature extraction model can also identify abnormal situations in the dataset where the traffic pattern deviates from the normal. For instance, on November 2, 2022, as depicted in Fig. 17, the identification accuracy (Top 1%) drops significantly to only 32%, while it surpasses 90% for other days. Fig. 19 visualizes the average traffic volume of all detectors over time for that day and the previous and following weeks. It is evident that the traffic volume changes follow a periodic pattern, but on November 2, 2022, a different trend emerged. Fig. 20 shows the traffic volume trends for two randomly selected detectors on these days. Research on relevant news revealed that the Hong Kong Observatory issued a Typhoon Signal No. 8 Northwest Gale or Storm warning at 1:40 PM that day. This explains why the traffic volume deviated from the regular pattern, with a decrease in the number of vehicles on the road. It can be seen that the daily feature extraction model can identify special days exhibiting divergent traffic patterns compared to regular days. In the future, integrating the proposed daily feature extraction model with reports on different events would enhance the analysis of traffic changes in various scenarios.

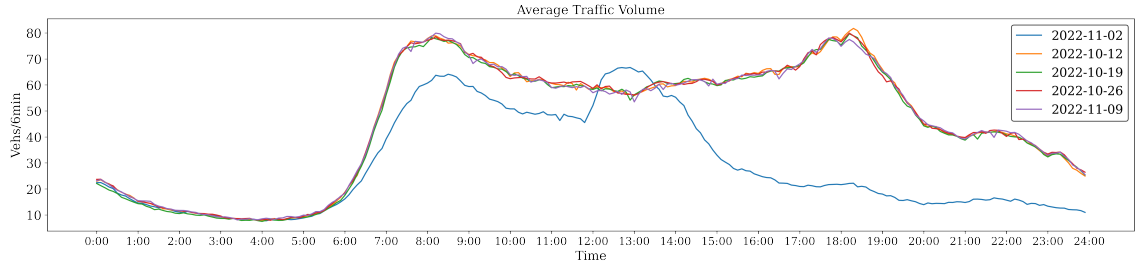


Fig. 19 Visualization of average traffic volumes of all detectors for few days

Furthermore, the proposed profile model can serve as one of the metrics to identify drifting traffic volumes for specific detectors. If a daily volume is inputted into the profile model and the correct detector ID does not rank within the top 1% of the output probability ranking, this daily volume likely deviates from its regular traffic pattern. For example, for detector ‘AID01117 lane1’, there are two days in the test set where

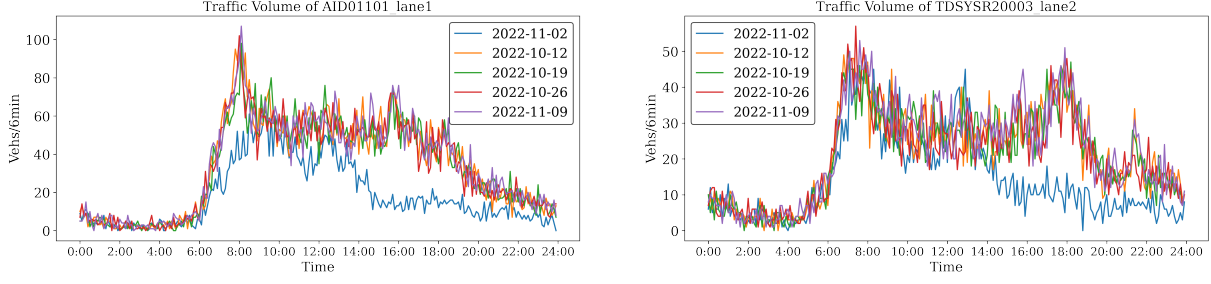


Fig. 20 Visualization of traffic volumes of two detectors of few days

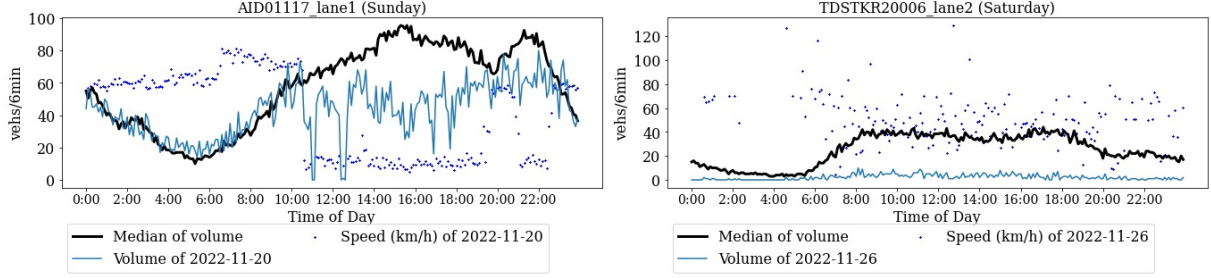


Fig. 21 Days diverging from typical traffic volume

it cannot be correctly identified. As mentioned earlier, one of those days is November 2nd, and the other is November 20th, 2022, as shown in Fig. 21. The abnormal days the profile model identifies differ from the usual traffic volume. Similarly, another detector identified different traffic patterns than usual on November 26th, 2022. Hence, the profile can be utilized as one of the indicators for identifying anomalies in future work. The deviation of traffic patterns from the norm can be attributed to various factors, such as events and detector malfunctions. For instance, on the left of Fig. 21, besides abnormal traffic volume, there is an extended period of abnormally low speed, which could result from detector malfunction. By examining the subsequent day, it was observed that the traffic speed in that lane returned to the normal range, suggesting that the anomaly on that particular day was likely caused by an event. In future work, the proposed profiles, along with traffic incidents, weather data, and other datasets, can be combined to provide a better analysis of the causes of traffic anomalies.

5 Conclusion

This paper proposes an approach for predicting traffic volume and speed on major strategic routes in Hong Kong. The proposed approach considers the reliability of detectors, allowing for accurate predictions with and without failed detectors.

The proposed approach employs two deep model structures: the daily feature extraction model and the traffic volume/speed prediction model. Additionally, it utilizes a machine learning technique, a hierarchical clustering algorithm. The daily feature extraction model extracts deep features from daily traffic volume to obtain

detectors' profiles. Detectors with similar profiles are grouped into clusters, and a deep prediction model is trained for each cluster.

Under ideal conditions where all detectors work, the MAE, RMSE, and MAPE for traffic volume prediction over the next 6 minutes are 5.17 vehicles/6 minutes, 7.64 vehicles/6 minutes, and 14.07%, respectively. Correspondingly, the MAE, RMSE, and MAPE values for traffic speed prediction are 3.70 km/h, 6.32 km/h, and 6.33%. The proposed method demonstrates its resilience to detector failures. In the presence of a 6% detector failure rate (equivalent to a dataset loss rate of 5.9%), the MAE, RMSE, and MAPE for traffic volume prediction are 5.24 vehicles/6 minutes, 7.81 vehicles/6 minutes, and 14.21%, respectively. The corresponding MAE, RMSE, and MAPE values for traffic speed prediction are 3.87 km/h, 6.55 km/h, and 6.68%.

Moreover, the suggested daily feature extraction model can be employed for detector classification and identification of special days within the dataset.

However, the proposed method is unsuitable for predicting traffic volume and speed in rare or unseen situations within the training dataset, such as non-recurrent congestion. To enhance the prediction performance of the proposed method in different scenarios, future research will integrate multiple-source data, such as traffic events, weather conditions, holiday schedules, and other relevant factors.

Declarations

Availability of data and materials. The dataset can be retrieved from data.gov.hk.

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